

CSE 4616

Lab 2: Simulating Wireless Channel Effects in OMNeT++

Duration: 1 hour 30 minutes — In-Lab Assessment

Tools: OMNeT++ and INET Framework

Objectives

- Connect the wireless channel theory from Lecture 1 to observable simulation behaviour in OMNeT++/INET.
- Understand how signal attenuation and path loss affect communication range and packet delivery.
- Observe the relationship between SNR, BER, and bitrate through rate adaptation in simulation.
- Simulate the effect of background noise and co-channel interference on wireless link quality.
- Extend the Lab 1 network topology with new wireless channel parameters and analyse the results.

Starting Point: This lab builds directly on the `WiredAndWirelessHostsWithAP` network from Lab 1. You will extend it by adding wireless channel parameters that model the real-world effects discussed in Lecture 1. No new devices are added to the NED file in Parts A–C; all changes are in the INI file. Part D adds one new wireless host to the NED file.

1. Theory Recap: From Lecture 1 to Simulation

1.1 Signal Attenuation and Path Loss

In Lecture 1 you learned that wireless signal strength decreases with distance. In free space, received power falls off as $1/d^2$, but in real environments (with walls, trees, and buildings) the decay is much steeper. This is modelled by a **path loss exponent** n :

$$P_r \propto \frac{P_t}{d^n}$$

where P_t is transmitted power, P_r is received power, and d is distance. In free space $n = 2$; in indoor environments n is typically 3–4.

In decibels, path loss between transmitter and receiver is:

$$\text{Path Loss (dB)} = P_{tx}(\text{dBm}) - P_{rx}(\text{dBm})$$

The **maximum tolerable path loss** is the difference between the transmitter power and the receiver sensitivity. When actual path loss exceeds this threshold, the receiver can no longer decode the signal and packets are lost.

Key parameters in INET that control path loss:

- `transmitter.power` — transmitted signal power (mW or dBm)
- `receiver.sensitivity` — minimum receivable power (dBm)
- `pathLoss.typename` — the mathematical model used (`FreeSpacePathLoss`, `TwoRayGroundReflection`, etc.)
- `pathLoss.alpha` — the path loss exponent n (default 2.0 for free space)

1.2 SNR, BER, and Rate Adaptation

From Lecture 1, the **Signal-to-Interference-and-Noise Ratio (SINR)** determines how easily a receiver can distinguish between a 0 and a 1:

$$\text{SINR} = \frac{S}{N + I}$$

A higher SINR means fewer bit errors (lower BER). The SINR–BER tradeoff depends on the modulation scheme: higher-order modulations (e.g., QAM256) achieve higher data rates at the same SINR but have a higher BER than simpler modulations (e.g., BPSK).

Rate adaptation means the transmitter dynamically selects the modulation and coding scheme that maximises data rate while keeping BER acceptable. As a device moves away from the access point and SINR drops, the link falls back to a lower but more robust data rate.

Key parameters in INET that control SNR and noise:

- `backgroundNoise.power` — thermal/environmental noise floor (dBm); increasing this lowers SNR
- `receiver.sensitivity` — minimum decodable power (dBm); sets the effective SNR floor
- `transmitter.bitrate` — the PHY data rate; in a full rate-adaptive model INET selects this automatically
- `receiver.energyDetection` — threshold below which the channel is considered idle (affects carrier sensing)

1.3 Multipath Fading and Environmental Obstructions

In Lecture 1 you saw that radio waves reach the receiver via multiple paths (reflection, diffraction, scattering), arriving at slightly different times. The constructive and destructive interference of these copies causes **fading**: random, time-varying fluctuations in received signal strength on top of the average path loss trend.

INET models this through the choice of path loss model and propagation medium parameters. The `TwoRayGroundReflection` model captures a ground-reflected path in addition to the line-of-sight path, which is a simple but realistic multipath model for outdoor scenarios.

Key parameters in INET that model multipath/environment:

- `pathLoss.typename = "TwoRayGroundReflection"` — models both direct and ground-reflected paths
- `pathLoss.typename = "FreeSpacePathLoss"` — ideal line-of-sight only (no reflections)
- `backgroundNoise.power` — also absorbs ambient multipath noise contributions

1.4 How INET Computes Packet Reception

When `wirelessHost1` sends a packet to the `accessPoint`, INET performs the following chain internally:

1. The **transmitter** generates a signal at `transmitter.power` dBm.
2. The **path loss model** reduces the signal power based on distance and the path loss exponent.
3. The **background noise** is added at the receiver.
4. SINR is computed: $\text{SINR} = P_{\text{received}} / P_{\text{noise}}$.
5. If $\text{SINR} \geq$ the threshold for the chosen modulation, the packet is decoded successfully. Otherwise it is dropped (packet error).

This chain directly maps to the Lecture 1 diagram: *Transmitted power* $\rightarrow$ *Propagation* $\rightarrow$ *Attenuation* $\rightarrow$ *Received Signal* $\rightarrow$ *SNR* $\rightarrow$ *BER* $\rightarrow$ *Packet error or success*.

2. Part A: Extending the Lab 1 NED File

File: `WiredAndWirelessHostsWithAP.ned` (from Lab 1 — **no changes needed**)

The NED file from Lab 1 is reused without modification for Parts A–C. The network structure (3 hosts, 1 router, 1 access point, 1 radio medium) is sufficient to observe all three channel effects. All changes are made exclusively in the INI file.

For Part D (Task 4), you will add one additional wireless host to the NED file to create an interference scenario.

Reminder of the Lab 1 topology:

```
wirelessHost1
  |
  (Wi-Fi)
  |
accessPoint ===== router ===== wiredHost2
  ||
  (Eth100M)
  ||
wiredHost1
```

wirelessHost1 is the only device on the wireless channel and communicates with wiredHost1 and wiredHost2 via the access point.

3. Part B: INI File — Baseline with Explicit Channel Parameters

File: `omnetpp.ini`

Add the following as a new `[Config]` section below the existing Lab 1 configuration. This baseline makes all wireless channel parameters explicit so you can modify them in each task.

```
1 [Config ChannelBaseline]
2 network = WiredAndWirelessHostsWithAP
3 sim-time-limit = 200s
4
5 **.addDefaultRoutes = false
6
7 # --- Application setup (same as Lab 1) -----
8 **.*Host*.numApps = 3
9
10 **.*Host*.app[0].typename = "UdpEchoApp"
11 **.*Host*.app[0].localPort = 1000
12
13 **.*Host*.app[*].typename = "UdpBasicApp"
14
15 **.[wiredHost1.app][1].destAddresses = "wirelessHost1"
16 **.[wiredHost1.app][2].destAddresses = "wiredHost2"
17 **.[wiredHost2.app][1].destAddresses = "wirelessHost1"
18 **.[wiredHost2.app][2].destAddresses = "wiredHost1"
```

```

19 **.wirelessHost1.app[1].destAddresses = "wiredHost1"
20 **.wirelessHost1.app[2].destAddresses = "wiredHost2"
21
22 **.Host*.app[1..].destPort          = 1000
23 **.Host*.app[1..].messageLength     = 512B
24 **.Host*.app[1..].sendInterval      = 0.5s
25 **.Host*.app[1..].stopTime          = 180s
26
27 # --- Wireless PHY: transmission power and sensitivity -----
28 **.wlan[0].radio.transmitter.power   = 20mW      # ~13 dBm
29 **.wlan[0].radio.transmitter.bitrate = 54Mbps    # 802.11g
    max
30 **.wlan[0].radio.receiver.sensitivity = -85dBm   # min.
    detectable power
31 **.wlan[0].radio.receiver.energyDetection = -85dBm # channel
    busy threshold
32 **.wlan[0].radio.carrierFrequency      = 2.4GHz
33
34 # --- Path loss model: free space (ideal, no reflections) -----
35 **.radioMedium.pathLoss.typename = "FreeSpacePathLoss"
36
37 # --- Background noise: thermal noise floor -----
38 **.radioMedium.backgroundNoise.power = -110dBm  # low noise
    baseline
39
40 # --- Visual positioning (keeps Lab 1 layout) -----
41 **.initialZ = 0m

```

Listing 1: Baseline configuration with explicit channel parameters

3.1 What Each New Parameter Does

3.2 Computing Maximum Range from These Parameters

With `transmitter.power = 20mW` (≈ 13 dBm) and `receiver.sensitivity = -85dBm`:

$$\text{Maximum Tolerable Path Loss} = 13 \text{ dBm} - (-85 \text{ dBm}) = 98 \text{ dB}$$

For free space path loss at $f = 2.4$ GHz:

$$\text{Path Loss (dB)} = 20 \log_{10}(d) + 20 \log_{10}(f) + 20 \log_{10}\left(\frac{4\pi}{c}\right) \approx 20 \log_{10}(d) + 100$$

Setting path loss = 98 dB and solving for d gives a maximum range of approximately **80 m** under free-space conditions.

Quick rule of thumb from Lecture 1: Every time distance doubles, free-space path loss increases by $20 \log_{10}(2) \approx 6$ dB. Every 10 dB reduction in transmit power halves the maximum range.

Parameter	Effect
<code>transmitter.power = 20mW</code>	Sets the transmitted signal power. Determines the maximum usable range before received power falls below sensitivity.
<code>receiver.sensitivity = -85dBm</code>	The weakest signal the radio can decode. Packets received below this power level are dropped regardless of noise.
<code>FreeSpacePathLoss</code>	Signal power falls off as $1/d^2$ with distance. Clean baseline with no reflections or multipath.
<code>backgroundNoise.power = -110dBm</code>	Constant noise floor added at every receiver. SINR = received signal power (dBm) – noise power (dBm).
<code>bitrate = 54Mbps</code>	PHY data rate. Higher rate requires higher SINR to achieve acceptable BER.

Table 1: New wireless channel parameters and their effects

4. Part C: Guided Parameter Explorations

The following three explorations correspond directly to the three Lecture 1 topics. Each one changes a small number of parameters so the effect is isolated and clearly observable.

4.1 Exploration 1: Signal Attenuation and Range

Path loss grows with distance. In INET, you can reduce the transmitter power to simulate a device that is effectively *farther away* from the access point (lower power means the signal behaves as if the range is shorter).

```

1 [Config Attenuation_Study]
2 extends = ChannelBaseline
3
4 # Reduce transmit power: the wireless link becomes marginal
5 # At 1mW (~0 dBm), max path loss = 0 - (-85) = 85 dB
6 # which corresponds to ~28m in free space
7 **wlan[0].radio.transmitter.power = 1mW
8
9 # Also try: 0.1mW = -10 dBm -> max path loss = 75 dB -> ~9m range
10 # **wlan[0].radio.transmitter.power = 0.1mW

```

Listing 2: Reduced transmit power to explore range limits

What to observe:

- At low power, packets from `wirelessHost1` may not reach the access point at all — you will see no echo traffic returning.
- The transition from “all packets delivered” to “no packets delivered” as power decreases illustrates the sharp reception threshold from Lecture 1 (sensitivity floor).

4.2 Exploration 2: SNR Degradation via Background Noise

SNR = received signal power – noise power (in dB). Increasing the noise floor directly reduces SNR, which increases BER and eventually causes packet loss even when the signal is strong enough to be detected.

```

1 [Config Noise_Study]
2 extends = ChannelBaseline
3
4 # Raise the noise floor progressively:
5 # Low noise (baseline):      -110 dBm  -> high SNR, low BER
6 # Medium noise:              -95 dBm   -> SNR reduced by 15 dB
7 # High noise:                -85 dBm   -> SNR ~ 0 dB at sensitivity
    -> most packets fail
8 **.radioMedium.backgroundNoise.power = -95dBm
9
10 # With transmitter.power = 20mW = 13 dBm, received power at short
11 # range (e.g., 5m) via free space ~ 13 - 60 = -47 dBm
12 # SNR = -47 - (-95) = 48 dB  -> still good
13 # SNR = -47 - (-85) = 38 dB  -> marginal for 54 Mbps

```

Listing 3: Increasing noise floor to degrade SNR

What to observe:

- As the noise floor rises toward the received signal level, the SNR margin shrinks.
- In Qtenv's console log, you will see an increasing number of packet reception failures as SNR falls.
- At very high noise (–85dBm), the noise equals the sensitivity, meaning even a clean signal at the sensitivity threshold cannot be decoded.

4.3 Exploration 3: Path Loss Model — Free Space vs. LogNormal Shadowing

Switching the path loss model from free space to **LogNormal Shadowing** introduces random variations in signal strength around the mean path loss. The model simulates signal fluctuations caused by obstacles, walls, and other environmental factors. The path loss exponent α controls the rate of signal decay: higher values correspond to more obstructed or indoor environments. For example, $\alpha = 3.5$ models moderate outdoor attenuation, while $\alpha = 4.5$ represents a heavily obstructed indoor environment with stronger signal variability.

```

1 [Config Multipath_Study]
2 extends = ChannelBaseline
3
4 # Option A: LogNormalShadowing (no NED change needed)
5 **.radioMedium.pathLoss.typename    = "LogNormalShadowing"
6 **.radioMedium.pathLoss.alpha       = 3.5
7
8 [Config Multipath_Indoor]
9 extends = ChannelBaseline
10 **.radioMedium.pathLoss.typename    = "LogNormalShadowing"
11 **.radioMedium.pathLoss.alpha       = 4.5    # heavy obstruction

```

Listing 4: Switching path loss model to simulate multipath

What to observe:

- With the LogNormal Shadowing model, signal strength exhibits random variations around the mean path loss, reflecting the impact of obstacles, walls, and environmental clutter.
- The path loss exponent α controls the rate of signal decay: higher α leads to faster attenuation and shorter effective range.
- This demonstrates the Lecture 1 point that *“receiving power is proportional to $1/d^2$ in free space, but can decay much faster in realistic environments due to shadowing and obstructions.”*

5. Part D: NED Extension — Adding an Interferer

File: WiredAndWirelessHostsWithAP.ned (modified)

To simulate **co-channel interference** (a key Lecture 1 topic), add a second wireless host (interferer) that transmits continuously on the same channel, degrading the link quality of wirelessHost1.

5.1 NED Change: Add the Interferer

Open WiredAndWirelessHostsWithAP.ned and add one submodule inside the existing submodules block:

```

1 // Add this submodule inside the existing submodules block
2 // Place it close to wirelessHost1 to maximise interference
3 interferer: WirelessHost {
4     parameters:
5         @display("p=62,140"); // positioned below wirelessHost1
6 }

```

Listing 5: Add interferer wireless host to the NED file

No new connections are needed — the interferer communicates wirelessly via the existing radioMedium.

5.2 INI Configuration for the Interference Experiment

```

1 [Config Interference_Study]
2 network = WiredAndWirelessHostsWithAP
3 sim-time-limit = 200s
4 **.addDefaultRoutes = false
5
6 # --- Original hosts: same as baseline -----
7 **.*Host*.numApps = 2
8 **.*Host*.app[0].typename = "UdpEchoApp"
9 **.*Host*.app[0].localPort = 1000
10 **.*Host*.app[*].typename = "UdpBasicApp"
11
12 **.[wiredHost1.app][1].destAddresses = "wirelessHost1"
13 **.[wirelessHost1.app][1].destAddresses = "wiredHost1"

```



```

14 **.wiredHost2.app[1].destAddresses      = "wiredHost1"
15
16 **.*Host*.app[1].destPort               = 1000
17 **.*Host*.app[1].messageLength          = 512B
18 **.*Host*.app[1].sendInterval           = 0.5s
19 **.*Host*.app[1].stopTime               = 180s
20
21 # --- Interferer: sends high-rate traffic on same channel -----
22 **.interferer.numApps = 1
23 **.interferer.app[0].typename           = "UdpBasicApp"
24 **.interferer.app[0].destAddresses      = "wiredHost2"
25 **.interferer.app[0].destPort           = 1000
26 **.interferer.app[0].messageLength      = 1400B    # large frames = more
    air time
27 **.interferer.app[0].sendInterval       = 0.05s    # very high rate =
    heavy interference
28 **.interferer.app[0].startTime          = 10s      # starts after main
    traffic established
29 **.interferer.app[0].stopTime           = 150s     # stops before end to
    show recovery
30
31 # --- Wireless channel (same for all devices) -----
32 **.wlan[0].radio.transmitter.power      = 20mW
33 **.wlan[0].radio.transmitter.bitrate    = 54Mbps
34 **.wlan[0].radio.receiver.sensitivity   = -85dBm
35 **.wlan[0].radio.receiver.energyDetection = -85dBm
36 **.wlan[0].radio.carrierFrequency       = 2.4GHz
37 **.radioMedium.pathLoss.typename        = "FreeSpacePathLoss"
38 **.radioMedium.backgroundNoise.power    = -110dBm
39 **.initialZ = 0m

```

Listing 6: Interference experiment INI configuration

5.3 What the Interferer Demonstrates

The interferer host transmits large, frequent packets on the same 2.4 GHz channel as wirelessHost1. Because they share the same radioMedium, their signals overlap at the access point:

- **Before $t = 10$ s:** Only wirelessHost1 is active. Packets flow cleanly with high SNR.
- **$t = 10$ s to 150 s:** The interferer transmits heavily. The effective SINR at the access point drops because the denominator $N+I$ grows. More packets from wirelessHost1 fail to decode.
- **After $t = 150$ s:** Interferer stops. Traffic from wirelessHost1 recovers. This directly models the Lecture 1 co-channel interference scenario (e.g., a 2.4 GHz phone interfering with an 802.11b/g network).

6. Part E: Parameter Reference and Calculation Guide

6.1 Power Unit Conversion (from Lecture 1)

Power (mW)	Power (dBm)	Conversion	Notes
1000 mW	30 dBm	$10 \log_{10}(1000)$	1 W transmitter
100 mW	20 dBm	$10 \log_{10}(100)$	Max 802.11g allowed
20 mW	13 dBm	$10 \log_{10}(20)$	Typical AP
1 mW	0 dBm	$10 \log_{10}(1)$	Typical smartphone
0.1 mW	-10 dBm	$10 \log_{10}(0.1)$	Low-power IoT

Table 2: Power unit conversion reference

6.2 Maximum Range Calculation

Given transmit power P_{tx} (dBm), sensitivity P_{rx} (dBm), and carrier frequency f (Hz), the free-space maximum range is:

$$d_{\max} = \frac{c}{4\pi f} \cdot 10^{(P_{tx} - P_{rx}) / 20}$$

where $c = 3 \times 10^8$ m/s.

6.3 SNR Margin

$$\text{SNR (dB)} = P_{\text{received}}(\text{dBm}) - P_{\text{noise}}(\text{dBm})$$

Typical minimum SNR requirements by modulation:

Modulation	Data Rate (802.11g)	Min. SNR for BER $< 10^{-5}$
BPSK	6 Mbps	~ 5 dB
QPSK	12 Mbps	~ 8 dB
16-QAM	24 Mbps	~ 15 dB
64-QAM	54 Mbps	~ 25 dB

Table 3: Approximate SNR requirements by modulation (802.11g)

7. Debugging Tips

- **No packets from wirelessHost1 at all:** Transmit power may be too low for the canvas distance. Increase `transmitter.power` or check that `receiver.sensitivity` is not too high (less negative).
- **Parameter path error for radioMedium:** The exact parameter path varies by INET version. Right-click `radioMedium` in Qtenv → Open Details to find the correct path in your installation.
- **Interferer not causing visible effect:** Reduce `interferer.app[0].sendInterval` further (e.g., `0.01s`) to increase channel load, or move the interferer's `@display` position closer to the access point.
- **Both attenuation and noise effects at once:** Do not combine low power *and* high noise in the same run when learning — isolate one variable at a time for clear observations.